

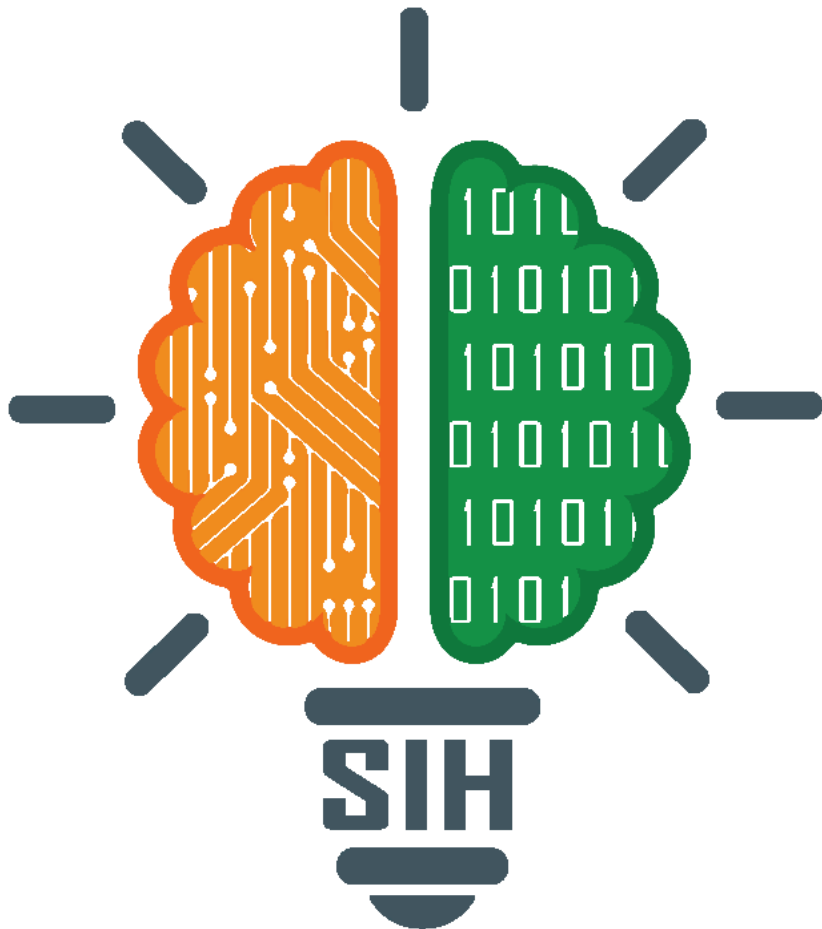
• Problem Statement ID –	SIH26086
• Problem Statement Title-	Hyperlocal Monsoon Onset & Break Prediction System (Block/Village Scale)
• Theme-	Agriculture, FoodTech & Rural Development
• PS Category-	Software
• Team ID-	120087
• Team Name-	TITANS

VARSHASETU

Hyperlocal Monsoon Onset, Break & Agricultural Advisory System

“Bridging Climate Intelligence with Every Farmer”

Ministry: Ministry of Earth Sciences (MoES) | Ref Org: NCMRWF
Institute: MVGR College of Engineering (Autonomous), Vizianagaram, AP



Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

How it addresses the problem

Innovation and uniqueness of the solution

Problem Context: The S2S Gap

- Coarse Synoptic Forecasts:** Operational models provide 50–100 km district forecasts that completely mask localized micro-climates and village-scale rainfall anomalies.
- Devastating Break Spells:** Sowing after pre-monsoon rain followed by an unpredicted 15-day dry break spell causes seedling desiccation and crippling re-sowing debt (₹5,000–₹8,000/acre).
- Input Leaching:** Heavy rainfall immediately post-fertilization washes expensive urea into waterways, causing financial devastation and ecological runoff.
- Missing Translation:** Meteorological outputs rarely translate raw millibars into actionable crop-specific guidance.

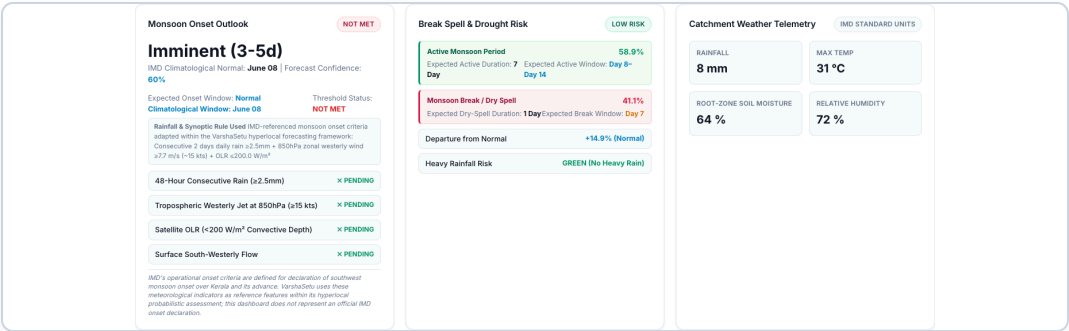
VarshaSetu Solution Architecture

- 7–30 Day S2S Outlook:** Probabilistic weekly rainfall departures, onset readiness, active spells, break spells (>7d continuous dry runs), and heavy rain alerts (>64.5 mm/day).
- IMD-Referenced Onset Evaluation:** Adapts 3 thermodynamic criteria: Rainfall (2d ≥2.5mm), 850 hPa wind (≥7.7 m/s), and OLR (≤200 W/m²). Status: **MET / NOT MET**.
- Dynamic ICAR-CRIDA Advisories:** Phenological stage rules for 6 Kharif crops (Paddy, Cotton, Soybean, Groundnut, Maize, Redgram) with moisture saturation thresholds.
- Contingent Crop Alternatives:** Automatic guidance to pivot to short-duration hardy cultivars (e.g. *Telangana Sona RNR 15048, Kadiri-6*) if onset is delayed >14 days.
- Inclusive Last-Mile Delivery:** 7 regional languages, Web Speech vernacular audio TTS, and sandboxed SMS/WhatsApp dispatches.

Uniqueness & Working Evidence

- Global Teleconnections → Hyperlocal Action:** Couples NOAA CPC ONI + IOD/MJO benchmarks with regional agro-meteorology.
- Dual Stakeholder Workflows:** Micro-level farmer advisories + macro-level extension officer mandal directives.
- Native Urdu RTL & Audio TTS:** True bi-directional layout transformation and voice readout for illiterate farmers.

LIVE PROTOTYPE EVIDENCE (STATUS GRID):



Working prototype status grid: Dharmasagar catchment showing adapted IMD onset criteria (MET), Active spell probability (74%), and Break spell risk (18%).

- ◆ **Predefined Pointers**
- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

🖥️ Technologies to be Used

Layer	Technology & Role
Frontend	HTML5, Vanilla CSS3 (light mode), JS ES2022, Leaflet.js 1.9.4 (GIS mapping), Chart.js 4.4.1 (rainfall/moisture trajectories).
Backend & APIs	Python 3.14 / Flask 3.1.0 microframework providing 10 modular REST API endpoints (CORS-enabled, structured JSON error envelopes).
AI / ML Engine	Scikit-Learn 1.6.1 , Joblib 1.4.2, NumPy 2.2.3. Serialized Physics-Informed Random Forest Classifier (break_model_rf_real_candidate.joblib).
Data & Connectors	NOAA CPC live connector (ENSO ONI), NASA POWER API (surface agro-met), Tenali observed records (2015–2025; 4,018 rows), Climatological benchmarks.
Persistence	SQLite3 serverless relational database for notification audit history (notification_audit table) and local telemetry caching.

Accessibility & i18n **W3C Web Speech API** (BCP-47 localized TTS), centralized JSON i18n engine (7 languages, 231 keys), native Urdu RTL support.

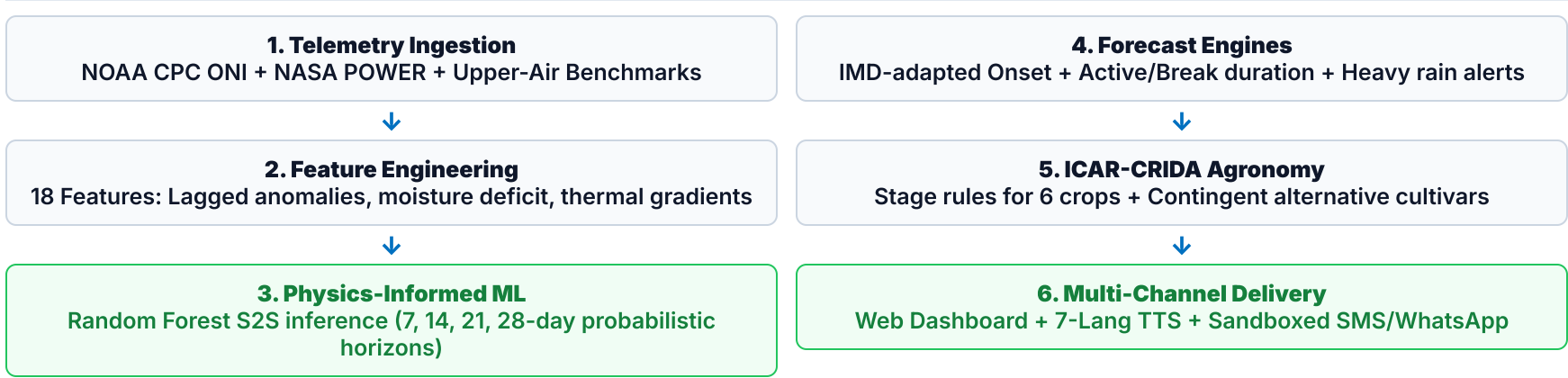
VERIFIED CODEBASE METRICS (PHYSICALLY PRESENT):

✓ 60/60 Pytest Tests Passed

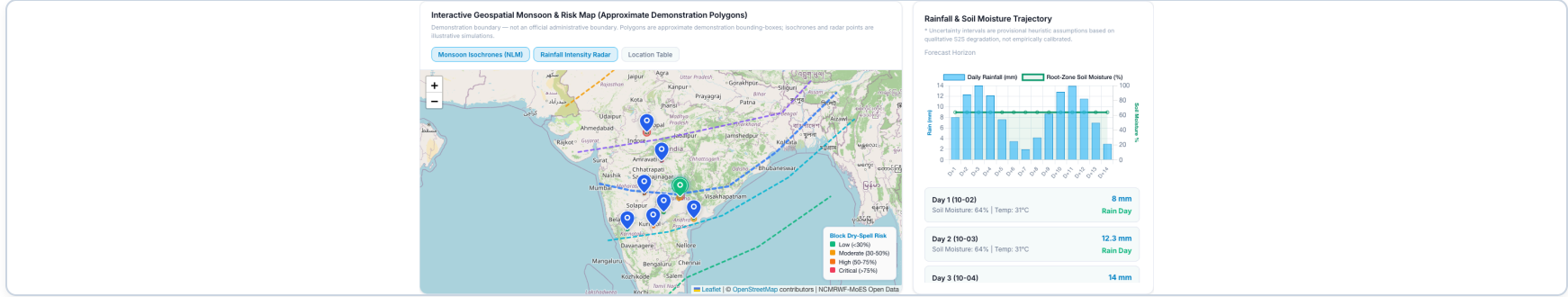
✓ 42/42 Mobile Viewports Verified

✓ 7 Languages (231 Keys)

⚙️ Methodology & Implementation Process



GEOSPATIAL RISK MAP & ISOCHRONE LAYER (PROTOTYPE EVIDENCE):



Mandatory Notice: Demonstration boundaries — approximate visualization, not official administrative boundaries. Isochrones (NLM) and radar points are simulated layers.

- ◆ **Predefined Pointers**
- Analysis of the feasibility of the idea
- Potential challenges and risks
- Strategies for overcoming these challenges

✔ **Feasibility Analysis**

- **Fully Functional Working Prototype:** Not an abstract concept. Full-stack implementation verified with 10 live Flask REST API routes, ML model, and SQLite database.
- **Proven Scientific ML Skill:**
 - **Brier Score:** 0.171 (vs 0.203 climatology)
 - **Brier Skill Score (BSS):** +0.158 (+15.8%)
 - **ROC-AUC:** 0.756 | **Recall:** 0.692
- **Zero Temporal Data Leakage:** Strictly evaluated using chronological train/test splits (2015–2022 train vs 2023–2025 out-of-sample test).
- **Zero Cost Burden on Farmers:** Runs seamlessly in mobile browsers without demanding high-end hardware, heavy app downloads, or expensive sensors.

⚠ **Potential Challenges & Risks**

- **S2S Atmospheric Predictability Limits:** Beyond 10–14 days, atmospheric chaos increases forecast uncertainty, limiting deterministic accuracy.
- **Hyperlocal Observation Sparsity:** Village-level precision relies on dense automatic weather station (AWS) telemetry, which varies by state.
- **NCMRWF Operational Socket Boundary:** Live supercomputing feeds require institutional MoES security clearance and departmental network agreements.
- **Rural Connectivity & Literacy Gaps:** Remote farmland suffers from 2G/3G connectivity drops and illiterate farmer accessibility hurdles.
- **Telecom Carrier Infrastructure:** Commercial SMS/WhatsApp broadcasts require enterprise DLT registrations and gateway infrastructure.

🛡 **Strategies for Overcoming Challenges**

- **Explicit Probabilistic Spreads:** Communicates probability distributions and widening uncertainty bounds rather than false deterministic precision.
- **Defensive Benchmark Fallbacks:** Gracefully falls back to 30-year IMD climatological normal benchmarks during sensor outages; zero fake success toasts.
- **Modular Plug-and-Play Design:** Architecture engineered with abstract connectors ready to ingest live NCMRWF NEPS 12km ensemble feeds upon authorization.
- **Vernacular Voice & Offline Table:** Web Speech TTS audio narrates alerts; accessible HTML table fallback supports low-bandwidth 2G conditions.
- **Phased Telecom Integration:** Sandbox demonstration verifies payload generation; ready for state portal integration (e.g. Rythu Bharosa).

PRACTICAL PHASED SCALE-UP ROADMAP



FROM PROTOTYPE TO NATIONWIDE AGRO-METEOROLOGICAL DEPLOYMENT

- ◆ **Predefined Pointers**
- Potential impact on the target audience
- Benefits of the solution (social, economic, environmental, etc.)

👤 **Potential Impact on Target Audience**

1. **Smallholder & Marginal Farmers (600M+ Dependent Citizens)**

- **Actionable Vernacular Directives:** Plain-language instructions (Sow, Delay Sowing, Provide Life-Saving Irrigation, Hold Fertilizer) replace confusing raw meteorological charts.
- **Re-Sowing Capital Protection:** Timely 10–14 day advance warnings of mid-season break spells prevent premature seed mortality.
- **Illiteracy Inclusivity:** One-tap Web Speech vernacular audio narration allows non-literate farmers to listen to advisories in their local dialect.
- **Contingent Cultivar Security:** Automatic guidance to pivot to drought-tolerant varieties (e.g., *Telangana Sona*, *Kadiri-6*) saves the farming season when onset is delayed.

2. **Agricultural Extension Officers (AEOs / KVK Scientists)**

- **Mandal-Scale Spatial Oversight:** Interactive risk map displays composite vulnerability clusters (Low to Severe) across 10–20 Panchayats simultaneously.
- **Contingency Seed Mobilization:** Alerts officials to preposition short-duration buffer seed stocks ahead of severe prolonged break spells.
- **Input Supply Synchronization:** Directs Primary Agricultural Credit Societies (PACS) to withhold subsidized urea distribution before extreme rainfall events to prevent leaching.

🌐 **Triple Impact Pillars & Value Chain**

💰 **Economic**

Saves ₹5,000–₹8,000/acre in prevented re-sowing losses and fertilizer washout. Optimizes crop yields.

👤 **Social**

Mitigates agrarian climate distress. Democratizes high-end MoES climate science in 7 local languages.

🌱 **Environmental**

Prevents chemical fertilizer runoff into water bodies; conserves groundwater via demand irrigation.

FARM-LEVEL DECISION VALUE CHAIN:

Hyperlocal Intelligence

→

Sowing Timing

→

Contingent Resilience

→

Preserved Capital

Scalability Architecture: Pluggable ICAR-CRIDA crop profiles support any regional Kharif cultivar. Centralized i18n JSON files enable rapid dialect onboarding. Decoupled REST APIs scale across Panchayats.

🏆 60/60 Pytest Tests Passed

📱 42/42 Mobile Viewports Verified

🌐 7 Regional Languages Supported

🔑 231 Verified Locale Keys/Language

@SIH Idea submission- Template

5

📖 Authoritative Operational & Scientific References

- **India Meteorological Department (IMD):** Operational criteria for Southwest Monsoon onset declaration over Kerala, S2S extended range protocols, and agro-met advisories. mausam.imd.gov.in
- **Ministry of Earth Sciences (MoES) & NCMRWF:** National Centre for Medium Range Weather Forecasting Extended Range Prediction System (ERPS) reanalysis and model documentation. ncmrwf.gov.in
- **NOAA Climate Prediction Center (CPC):** Oceanic Niño Index (ONI) 3-month running mean SST anomalies in the Niño 3.4 region (Real external connector). cpc.ncep.noaa.gov
- **Australian Bureau of Meteorology (BOM):** Indian Ocean Dipole (DMI) and Wheeler-Hendon Real-time Multivariate MJO (RMM1/RMM2) convective wave tracking. bom.gov.au/climate
- **ICAR - CRIDA Protocols:** Indian Council of Agricultural Research & Central Research Institute for Dryland Agriculture technical contingency bulletins for rainfed Kharif crops. cida.in
- **NASA POWER Project:** Prediction of Worldwide Energy Resources surface agro-climatology data access for solar radiation, relative humidity, and surface temperature. power.larc.nasa.gov
- **WMO S2S Prediction Project:** Subseasonal-to-Seasonal prediction research, Brier Skill Score standards, and multi-model ensemble verification protocols. s2sprediction.net
- **W3C Web Speech & TRAI DLT Guidelines:** SpeechSynthesis BCP-47 localized voice standards & Telecom Regulatory Authority of India rural messaging template specifications.

PROTOTYPE READY FOR MOES / NCMRWF PILOT EVALUATION

VarshaSetu is an active, verified software prototype with 60/60 automated tests passed, persistent SQLite audit logging, and full cross-device verification.

SIH26086 COMPLIANT

SCIENTIFIC INTEGRITY, PROTOTYPE BOUNDARIES & ETHICAL DISCLOSURES:

- **NCMRWF Connectivity:** NCMRWF upper-air variables are represented via climatological benchmark fallback in prototype; live supercomputing integration is a post-hackathon deployment step.
- **IMD Onset Advisory:** IMD-referenced monsoon onset criteria adapted within the VarshaSetu hyperlocal forecasting framework; this dashboard does not issue an official IMD onset declaration.
- **Rural Messaging Gateway:** SMS/WhatsApp delivery workflow demonstrated through a sandboxed simulation gateway without live commercial telecom charges.
- **Cartographic Boundaries:** Demonstration boundaries are approximate visualization bounding-boxes, not official Survey of India administrative cadastral boundaries.